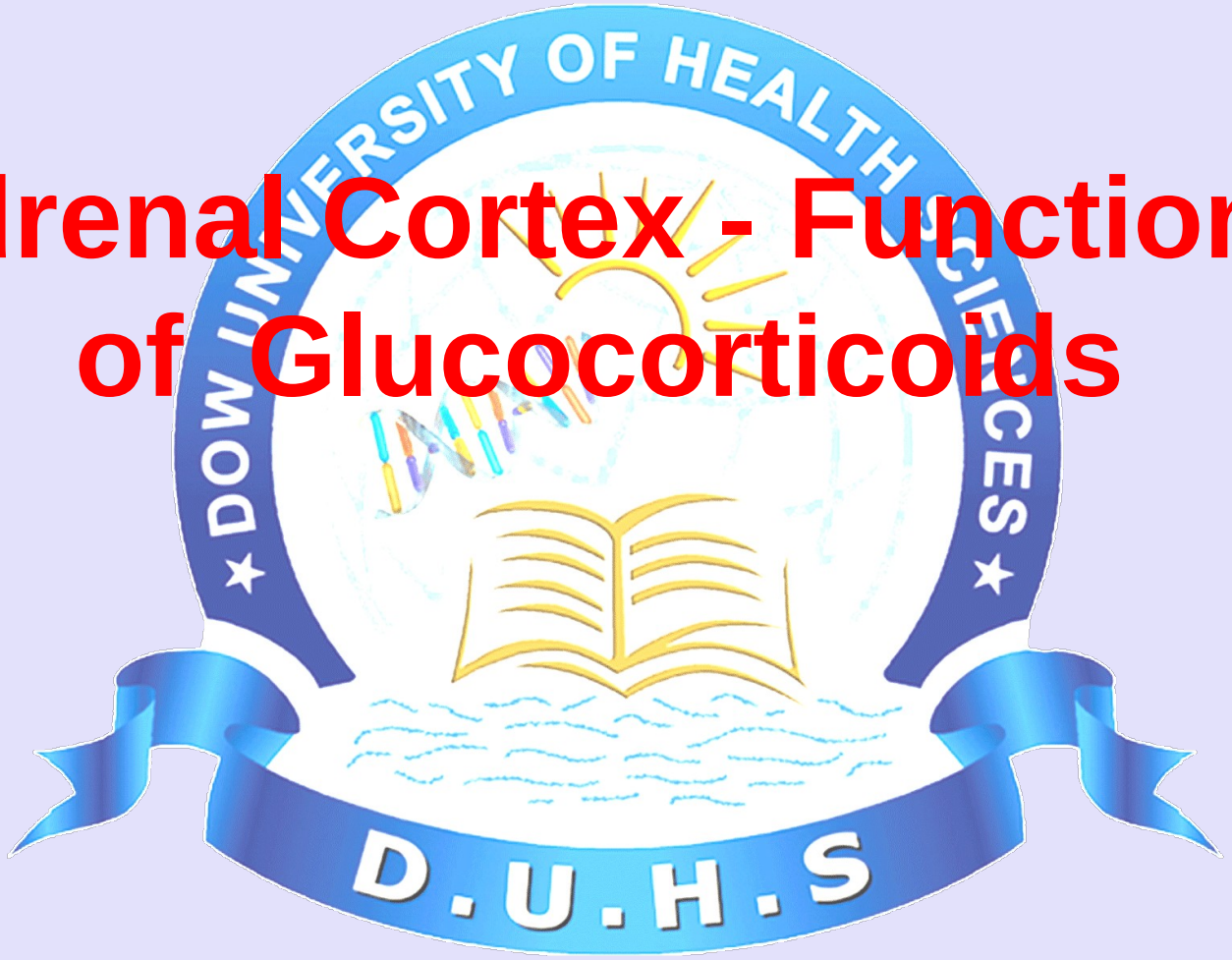
Adrenal Cortex - Functions of Glucocorticoids



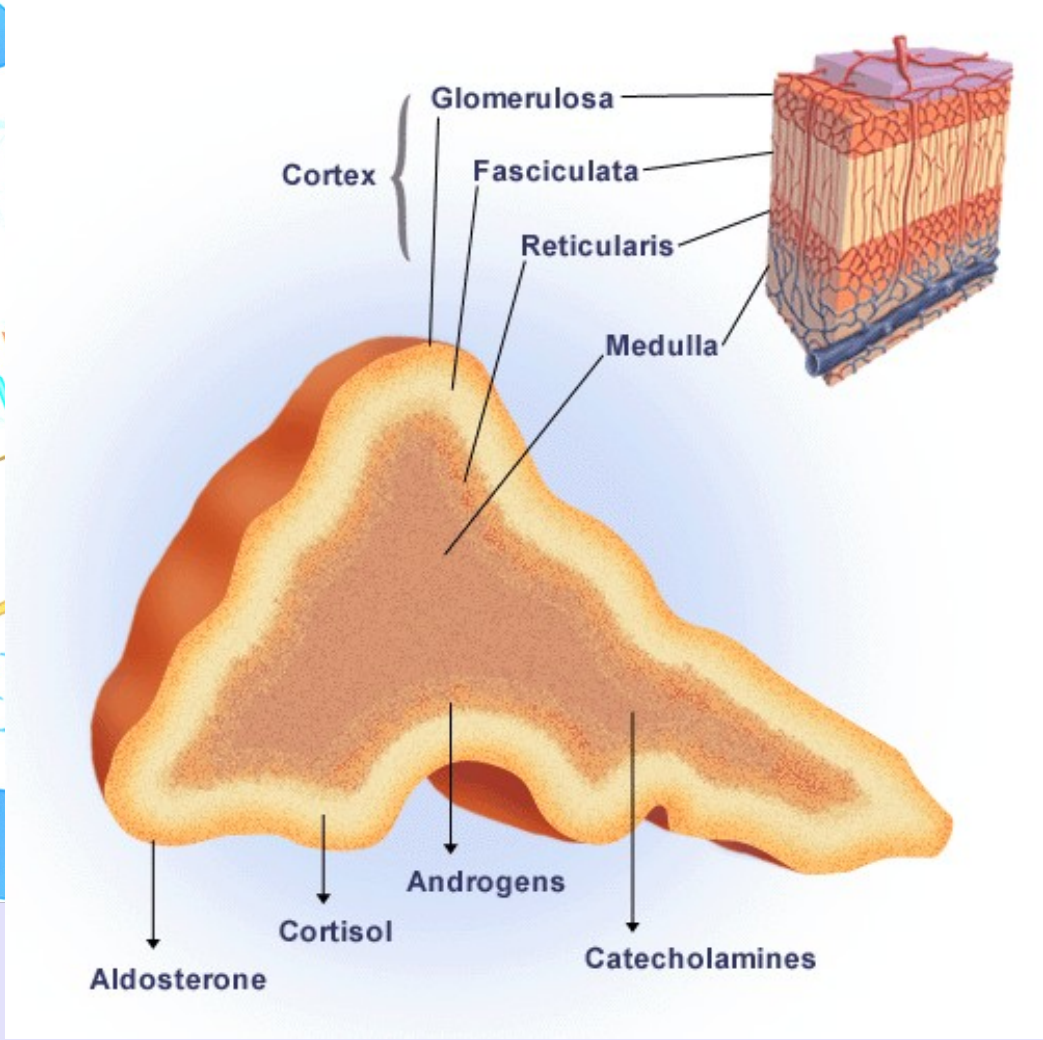
Learning Objectives

By the end of the lecture, the student should be able to :

- Give an overview of the hormones of Adrenal Cortex.
- Give the functions of Cortisol and its actions on :
  - Carbohydrate, protein and fat metabolism.
  - Anti-inflammatory action.
  - immunity and blood cells
  - in stress
  - allergic responses.

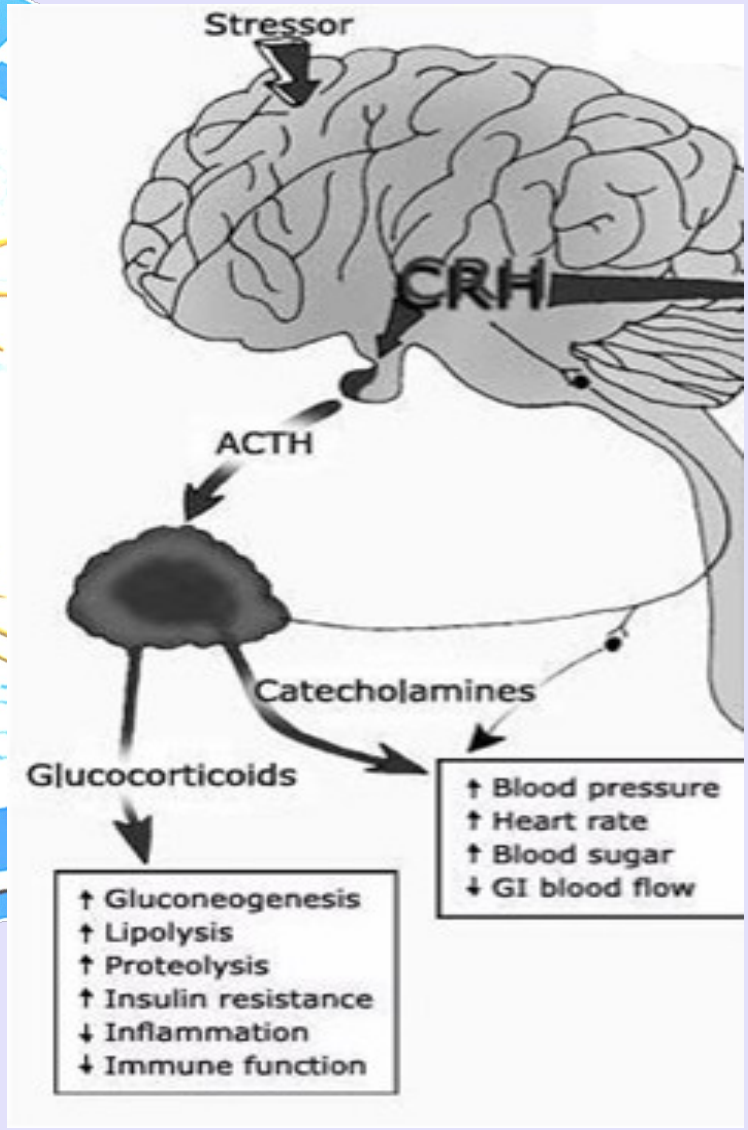
Hormones of adrenal cortex

- Glucocorticoid (cortisol and corticosterone)
- Mineralocorticoid (aldosterone)
- Androgens.



Physiological actions of cortisol

- Actions on carbohydrate metabolism
- Actions on protein metabolism
- Actions on fat metabolism
- Anti-inflammatory action
- Effects on immunity and blood cells
- Actions in stress
- Effects on allergic responses



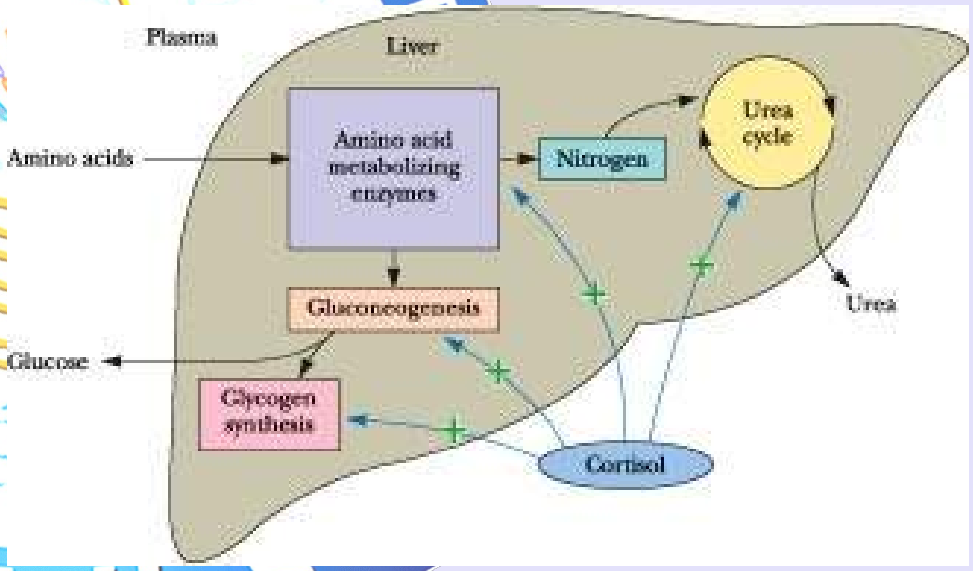
Effects of Cortisol on Carbohydrate Metabolism



Stimulation of Gluconeogenesis:

- Stimulate gluconeogenesis (formation of carbohydrate from proteins and some other substances) by the liver.
- Increasing gluconeogenesis as much as 6- to 10-fold.

- **Effects of cortisol on carbohydrate and protein metabolism in the liver.**



Decreased Glucose Utilization by Cells

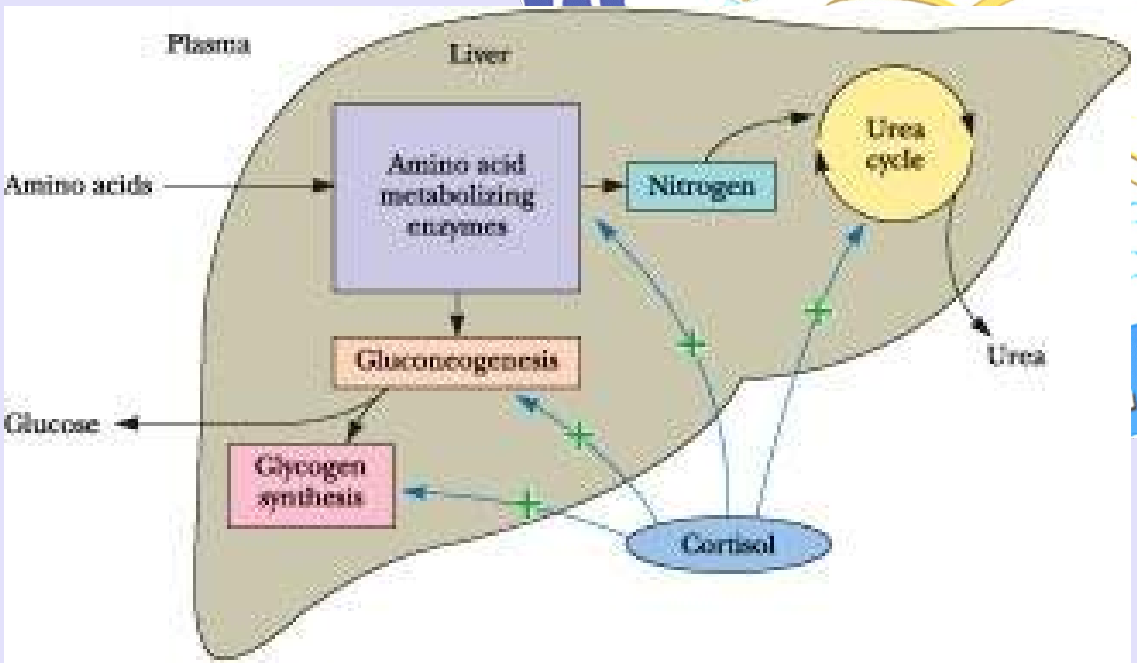
- Cortisol causes a moderate decrease in the rate of glucose utilization by most cells in the body.

Reduction in Cellular Protein

- Reduction of protein stores in essentially all body cells except those of liver.
- This is caused by both decreased protein synthesis and increased catabolism of protein already in the cells.
- In the presence of great excesses of cortisol, the muscles can become so weak that the person cannot rise from the squatting position.

Cortisol Increases Liver and Plasma Proteins

- Cortisol enhance amino acid transport into liver cells (but not into most other cells) and to enhance the liver enzymes required for protein synthesis.



Effects of cortisol on carbohydrate and protein metabolism in the liver.

Diminished Transport of Amino Acids into Extrahepatic Cells, and Enhanced Transport into Hepatic Cells

- Cortisol depresses amino acid transport into muscle cells and other extrahepatic cells.

- The decreased transport of amino acids into extrahepatic cells decreases their intracellular amino acid concentrations and consequently decreases the synthesis of protein.

Effects of Cortisol on Fat Metabolism

Mobilization of Fatty Acids

- Mobilization of fatty acids from adipose tissue
  - Decreased α glycerophosphate level which is required for deposition and maintenance of triglycerides
- Increases the concentration of free fatty acids in the plasma .

Mobilization of Fatty Acids

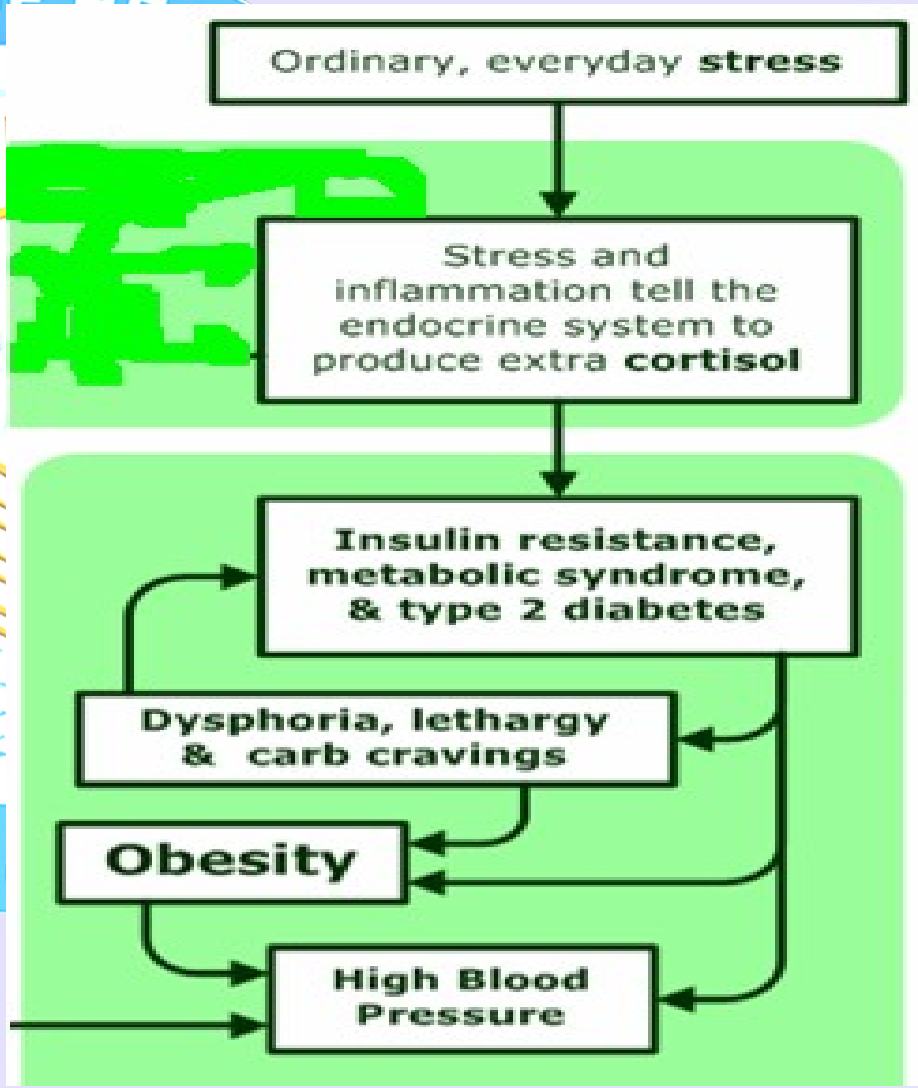
- Increases their utilization for energy
  - Shift from use of glucose for energy in stress
  - Long term conservation of glucose and glycogen
- Direct effect to enhance the oxidation of fatty acids in the cells.

Obesity Caused by Excess Cortisol

- Many people with excess cortisol secretion develop a peculiar type of obesity, with excess deposition of fat in the chest and head regions of the body, giving a buffalo-like torso and a rounded “moon face.”

Cortisol is Important in Resisting Stress and Inflammation

- Almost any type of stress, whether physical or neurogenic, causes an immediate and marked increase in ACTH secretion by the anterior pituitary gland, followed within minutes by greatly increased adrenocortical secretion of cortisol.



Types of stress that increase cortisol release

Some of the different types of stress that increase cortisol release are the following:

1. Trauma of almost any type
2. Infection
3. Intense heat or cold
4. Injection of norepinephrine and other sympathomimetic drugs
5. Surgery
6. Injection of necrotizing substances beneath the skin
7. Restraining an animal so that it cannot move
8. Almost any debilitating disease.

Anti-inflammatory Effects of High Levels of Cortisol

- It can block the early stages of the inflammation process before inflammation even begins,

or

- If inflammation has already begun, it causes rapid resolution of the inflammation and increased rapidity of healing.

Cortisol Prevents the Development of Inflammation by Stabilizing Lysosomes

- Cortisol stabilizes the lysosomal membranes.
- Cortisol decreases the permeability of the capillaries, probably as a secondary effect of the reduced release of proteolytic enzymes. This prevents loss of plasma into the tissues.
- Cortisol decreases both migration of white blood cells into the inflamed area and phagocytosis of the damaged cells.

Cortisol Prevents the Development of Inflammation

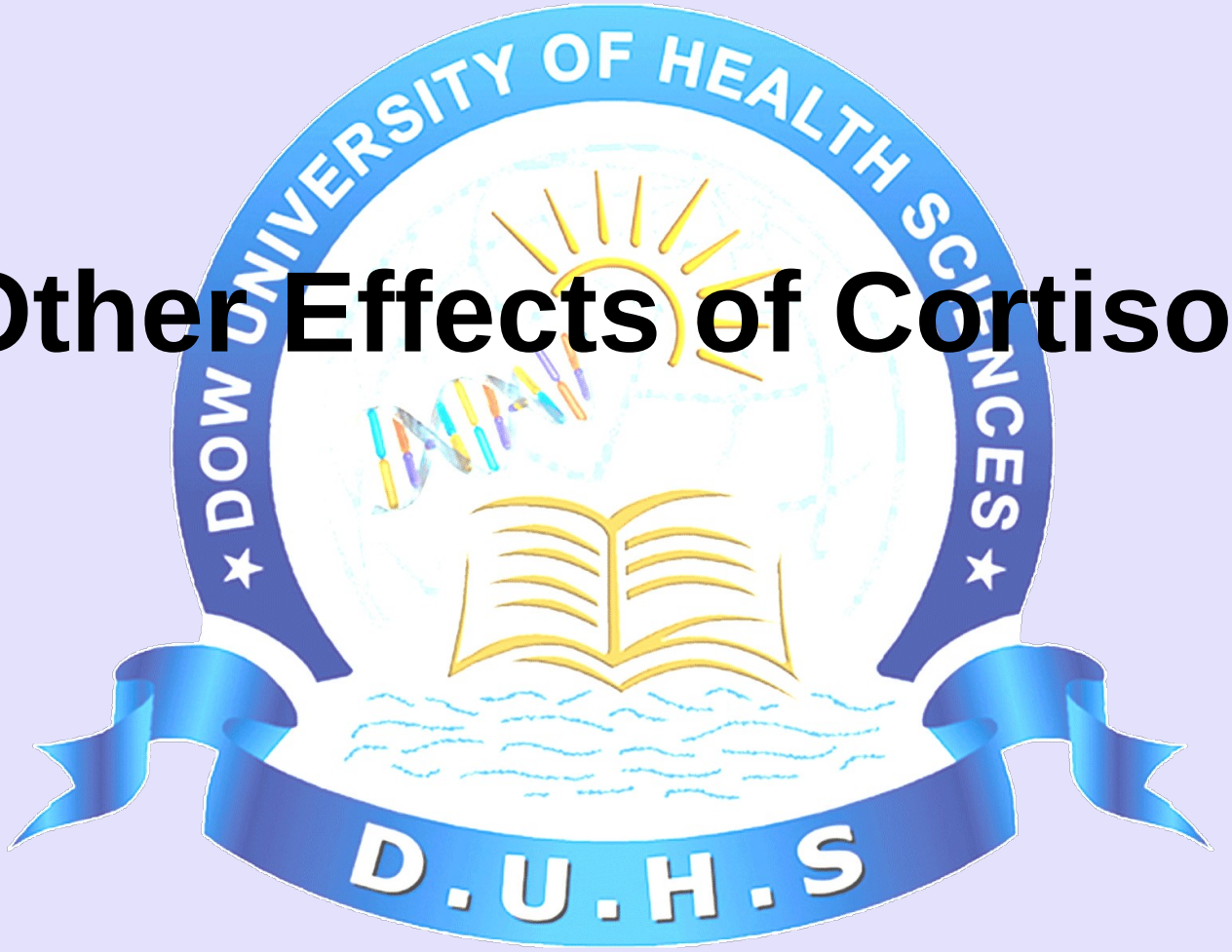
- Cortisol suppresses the immune system, causing lymphocyte reproduction to decrease markedly. The T lymphocytes are especially suppressed.

  Cortisol has an almost global effect in reducing all aspects of the inflammatory process.

Cortisol role in fever

- Cortisol attenuates fever mainly because it reduces the release of interleukin-1 from the white blood cells, which is one of the principal excitants to the hypothalamic temperature control system. The decreased temperature in turn reduces the degree of vasodilation.

Other Effects of Cortisol



Cortisol Blocks the Inflammatory Response to Allergic Reactions

- Cortisol effectively prevents shock or death in anaphylaxis, which otherwise kills many people.

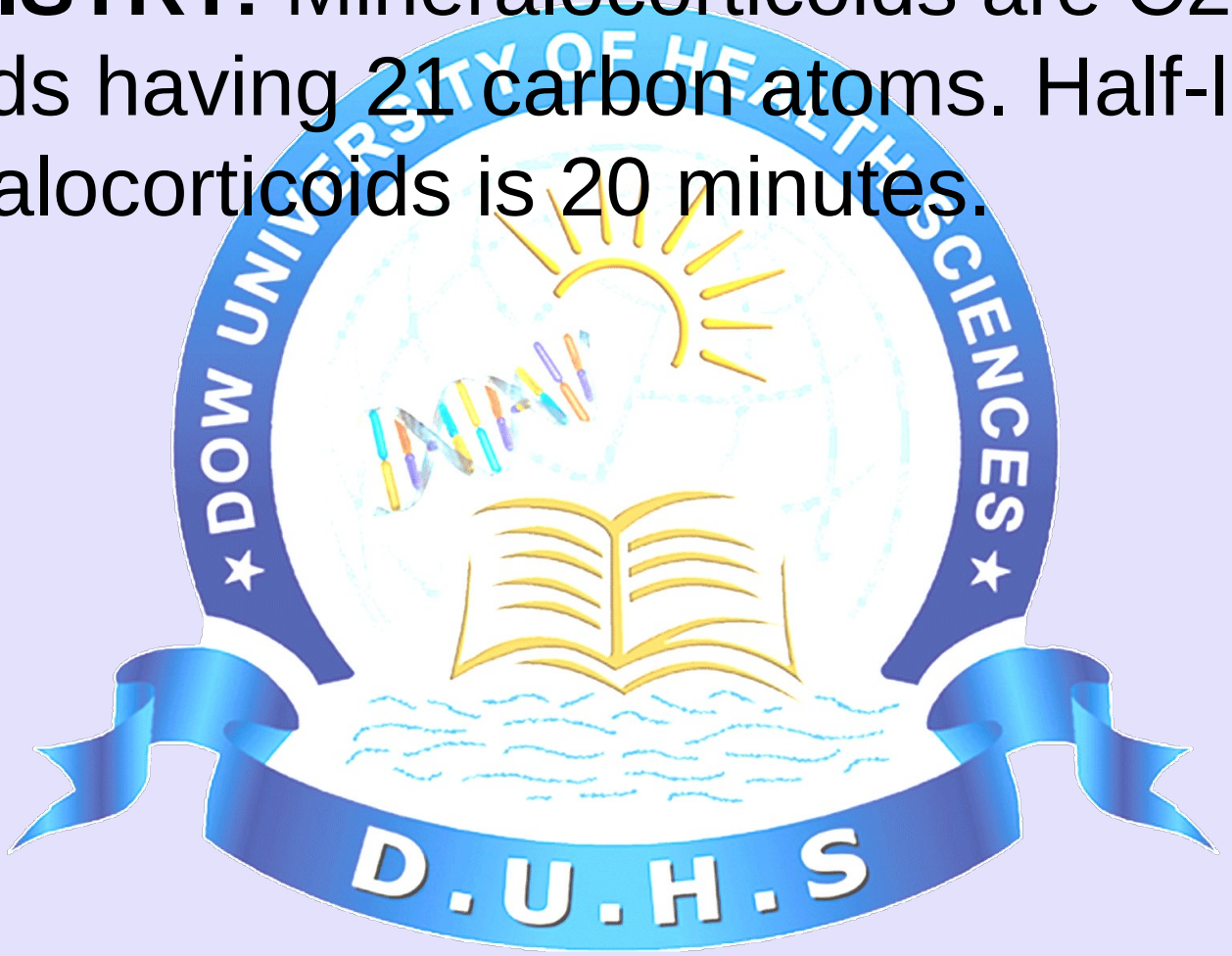
MINERALOCORTICOIDS

Mineralocorticoids are the corticosteroids that act on the minerals (electrolytes), particularly sodium and potassium.

1. Aldosterone
2. 11-deoxycorticosterone

SOURCE OF SECRETION: zona glomerulosa

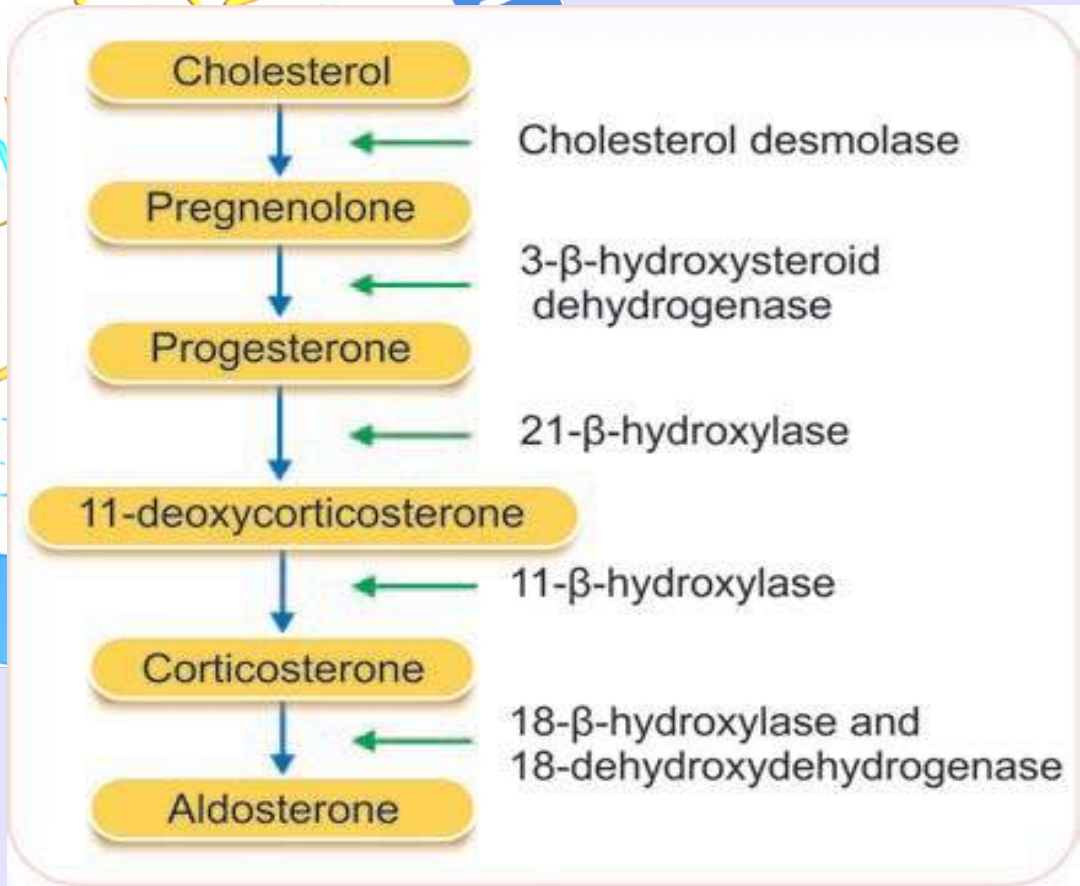
CHEMISTRY: Mineralocorticoids are C21 steroids having 21 carbon atoms. Half-life of mineralocorticoids is 20 minutes.



SYNTHESIS

All adrenocortical hormones are steroid in nature and are synthesized mainly from cholesterol that is absorbed directly from the circulating blood.

ALDOSTERONE



FUNCTIONS OF MINERALOCORTICOIDS

Aldosterone is very essential for life and it maintains the osmolarity and volume of ECF.

Aldosterone has three important functions.

It increases:

1. Reabsorption of sodium from renal tubules
2. Excretion of potassium through renal tubules
3. Secretion of hydrogen into renal tubules.

On Sodium Ions

Aldosterone increases the Sodium absorption and secretion of K+ in distal convoluted tubules of the kidney.

On Extracellular Fluid Volume

Water reabsorbed along with Sodium ions in equal amount ,

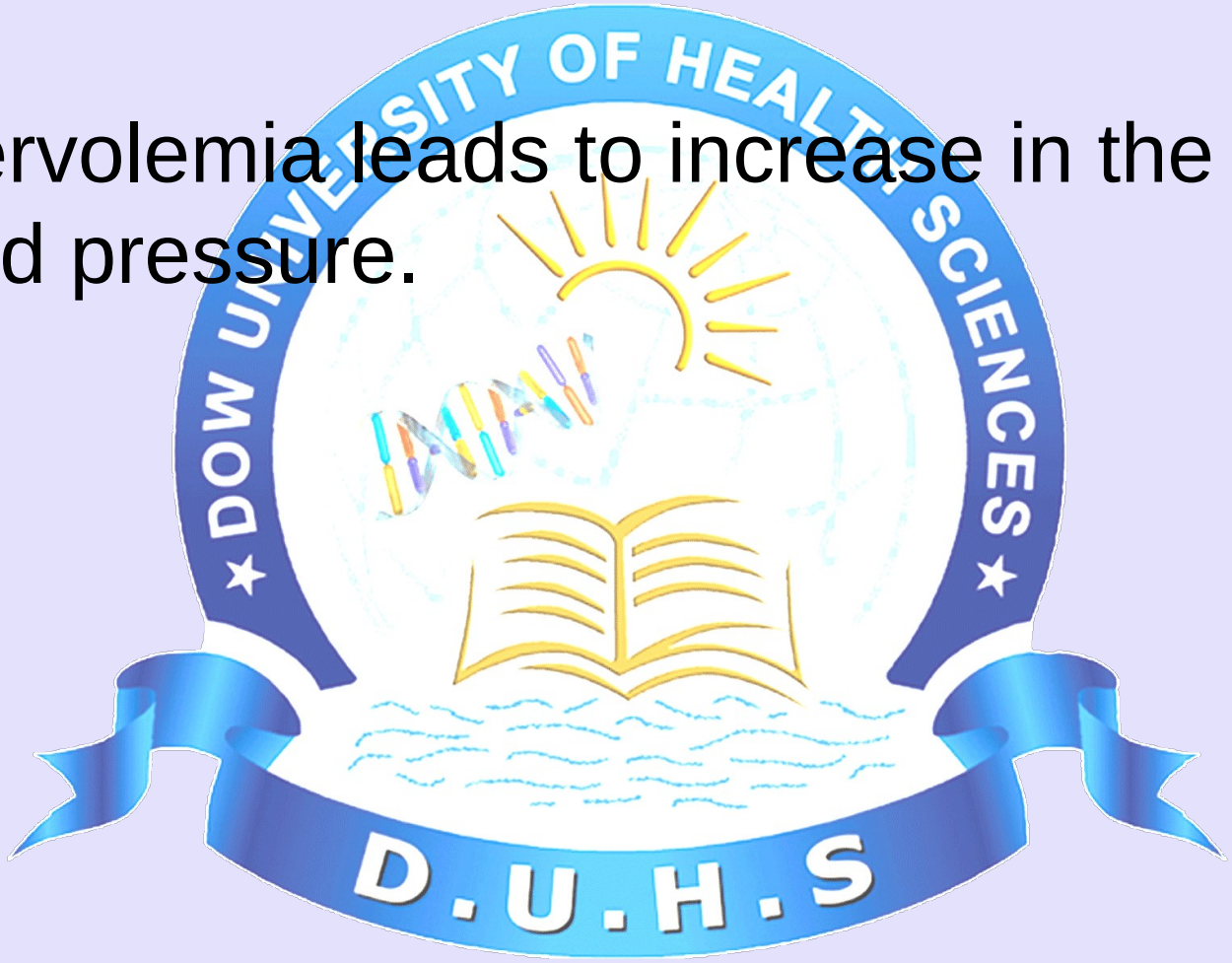
Increases the ECF volume

Mild hypernatremia.

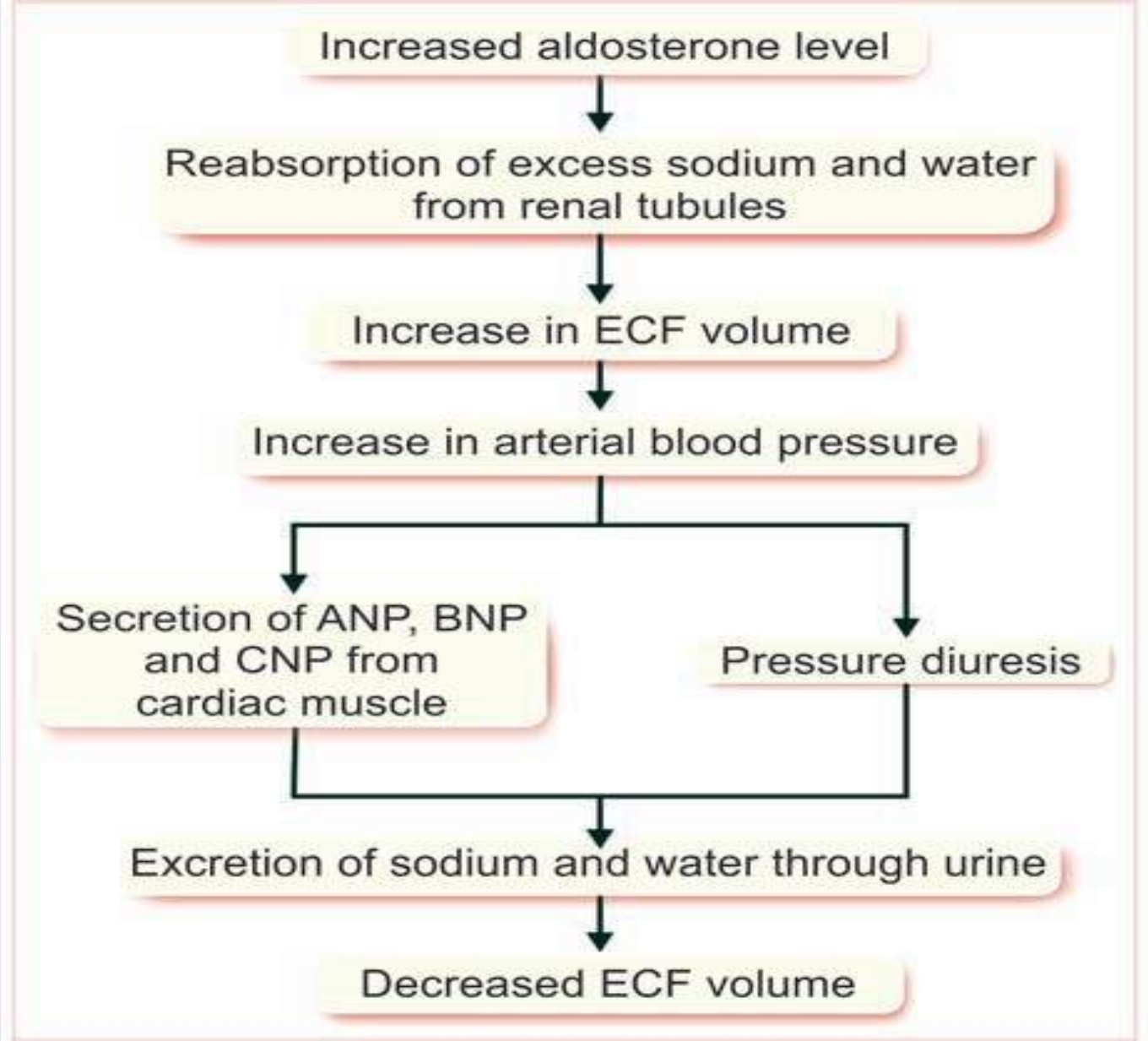
It induces thirst, leading to intake of water which again increases the ECF volume and blood volume.

On Blood Pressure

Hypervolemia leads to increase in the blood pressure.



Aldosterone escape



On Potassium Ions

Aldosterone increases the potassium excretion through the principle cells of renal collecting tubules.

On Hydrogen Ion Concentration:

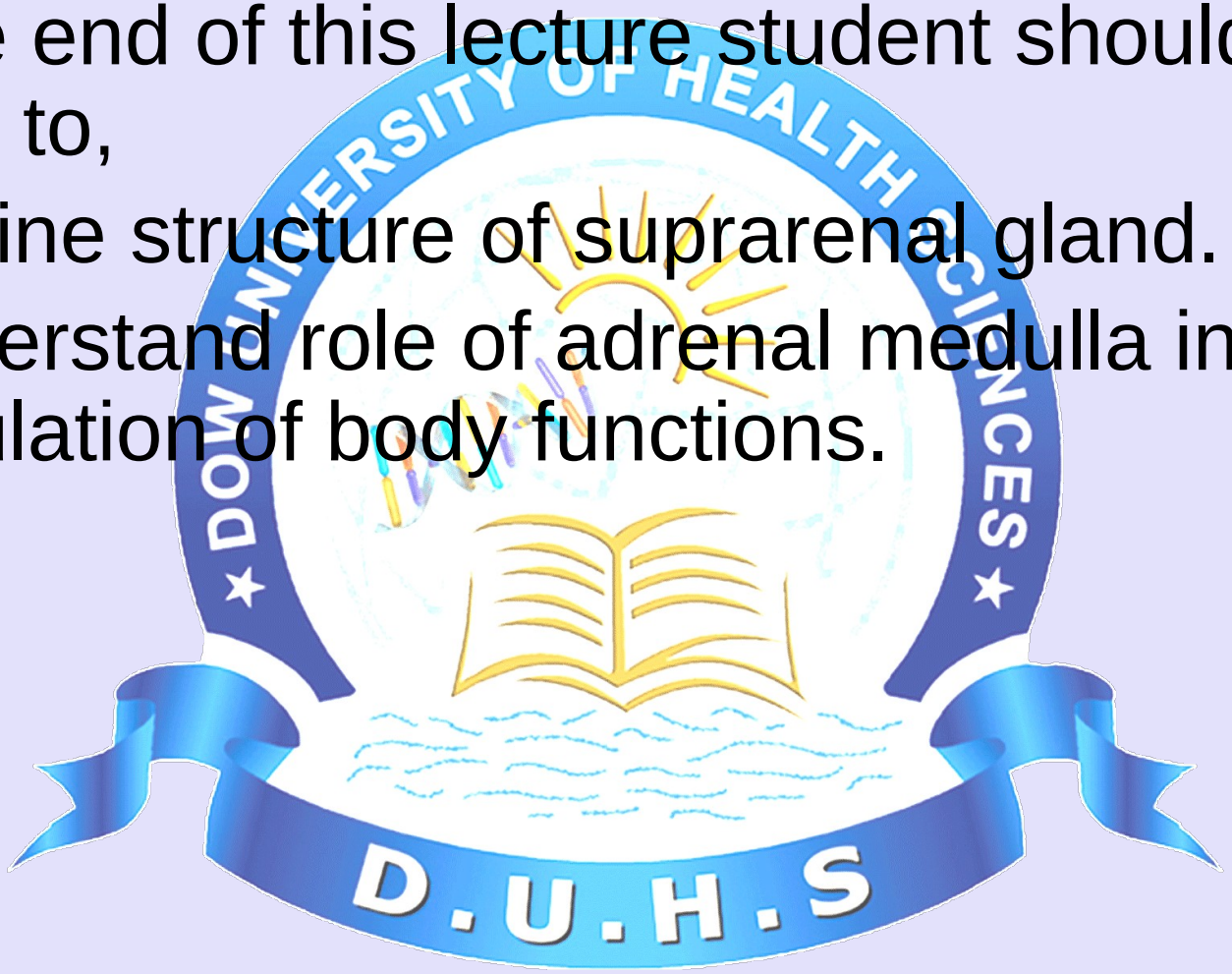
Secretion of hydrogen ions in exchange for sodium in the *intercalated cells* of the cortical collecting tubules.

ADRENAL MEDULLA SECRETION,MOA,AND DISORDER

LEARNING OBJECTIVES

By the end of this lecture student should be able to,

- Outline structure of suprarenal gland.
- Understand role of adrenal medulla in regulation of body functions.



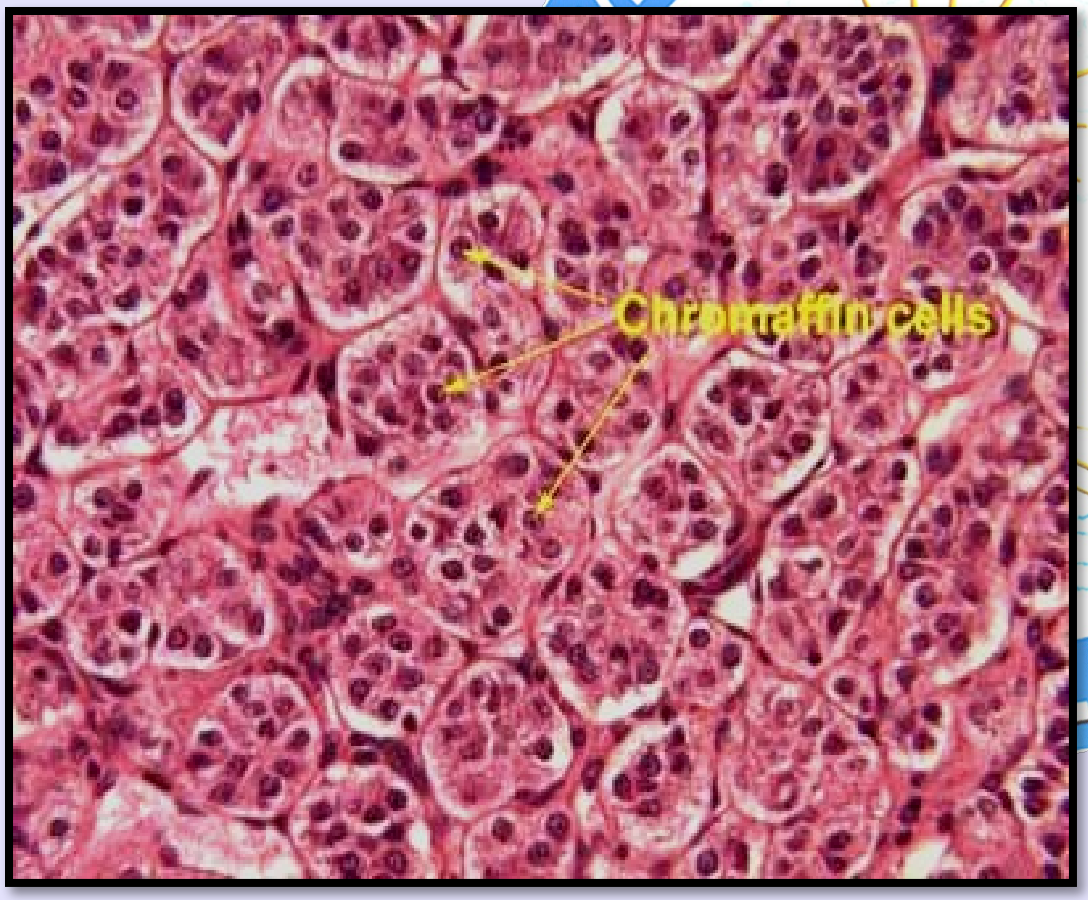
ADRENAL MEDULLA

- It secretes the catecholamines, adrenaline (80%) and noradrenaline (20%)
- They are called the fight or flight hormones or the 3Fs (fear, fight, flight)
- Opioid peptides are also synthesized and co secreted with the hormones
- They are released in response to stimulation by acetylcholine from the preganglionic sympathetic nerve endings

ADRENAL MEDULLA:

The principal cells of the medulla are the **chromaffin cells**. They secretes **adrenaline & noradrenaline**.

.





RELEASE OF THE ADRENAL MEDULLARY HORMONES

- Release of adrenaline and noradrenaline is triggered by nervous stimulation in response to physical or mental stress.
- The hormones bind to **adrenergic receptors**— transmembrane proteins in the plasma membrane of many cell types.

HORMONES OF THE ADRENAL MEDULLA

- Produces two similar hormones (catecholamines)
  - Epinephrine
  - Norepinephrine
- These hormones prepare the body to deal with short-term stress

FUNCTION OF ADRENAL MEDULLA

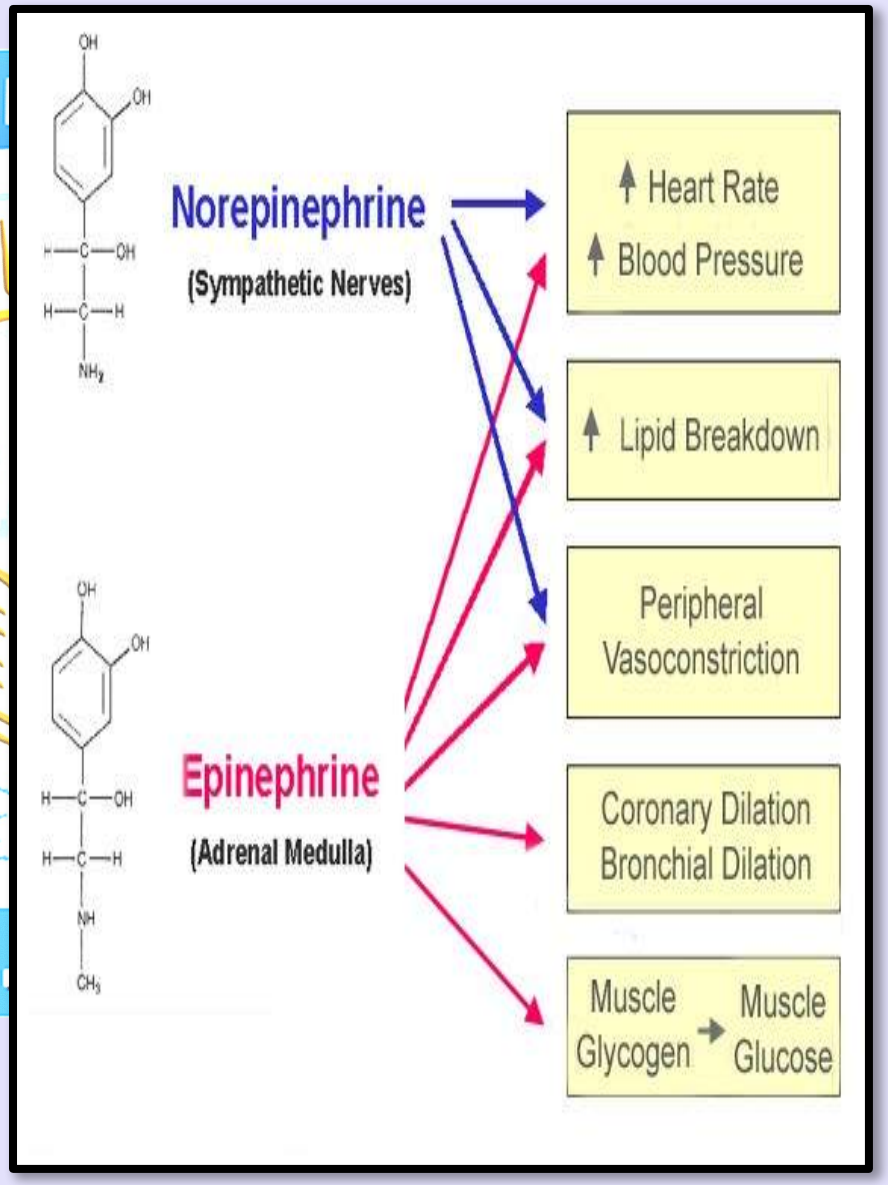
- The adrenal medulla is made up of chromaffin cells where in the amino acid tyrosine is converted to catecholamines like adrenaline, noradrenaline and dopamine.
- The function of adrenal medulla is governed by nervous system and the chromaffin cells release the hormones responsible for various metabolic reactions.
- The adrenal medullary hormones are involved in control of heart rate, blood pressure, blood vessel constriction in various organs like skin, gastrointestinal tract and also dilation of blood vessel in skeletal muscles.

Actions of Adrenal medullary hormones

- Increase blood glucose by:
  1. Promote glycogenolysis and gluconeogenesis in the liver
  2. Inhibit secretion of insulin
- Stimulate lipolysis with resultant increase in FFA
- Increase heart rate contraction and cause branchodilatation
- Increase constriction of visceral blood vessels and vasodilatation of skeletal muscles
- Decrease function of visceral organs
- Increase the basal metabolic rate and total oxygen consumption (calorigenic)

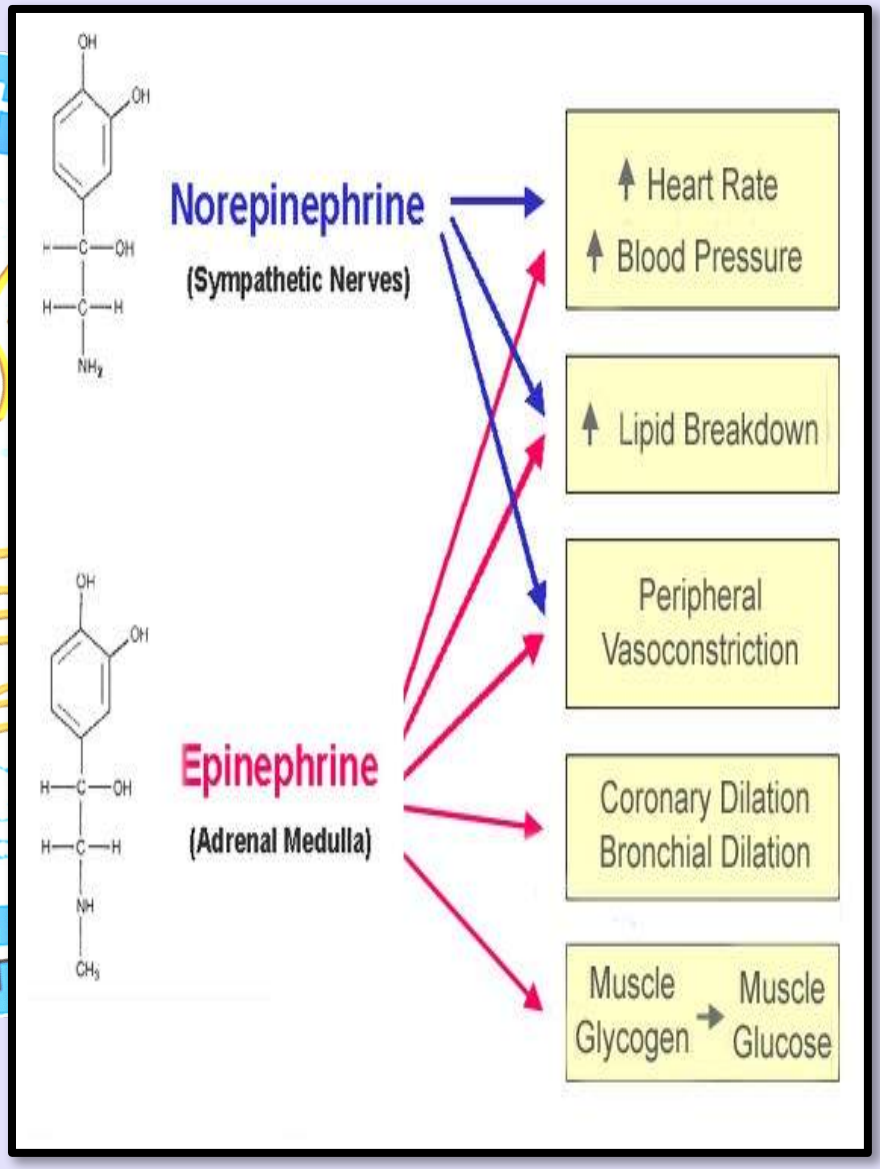
EFFECTS OF EPINEPHRINE

- Adrenaline, also known as epinephrine,
- is secreted from the adrenal medulla,
- boosting the supply of oxygen and glucose to the brain.
- increases heart rate and blood pressure,
- dilates pupils,
- constricts blood vessels in the skin and intestines,
- causing the hairs on your arms to stand on end.
- Adrenaline is a breakdown product of noradrenaline.



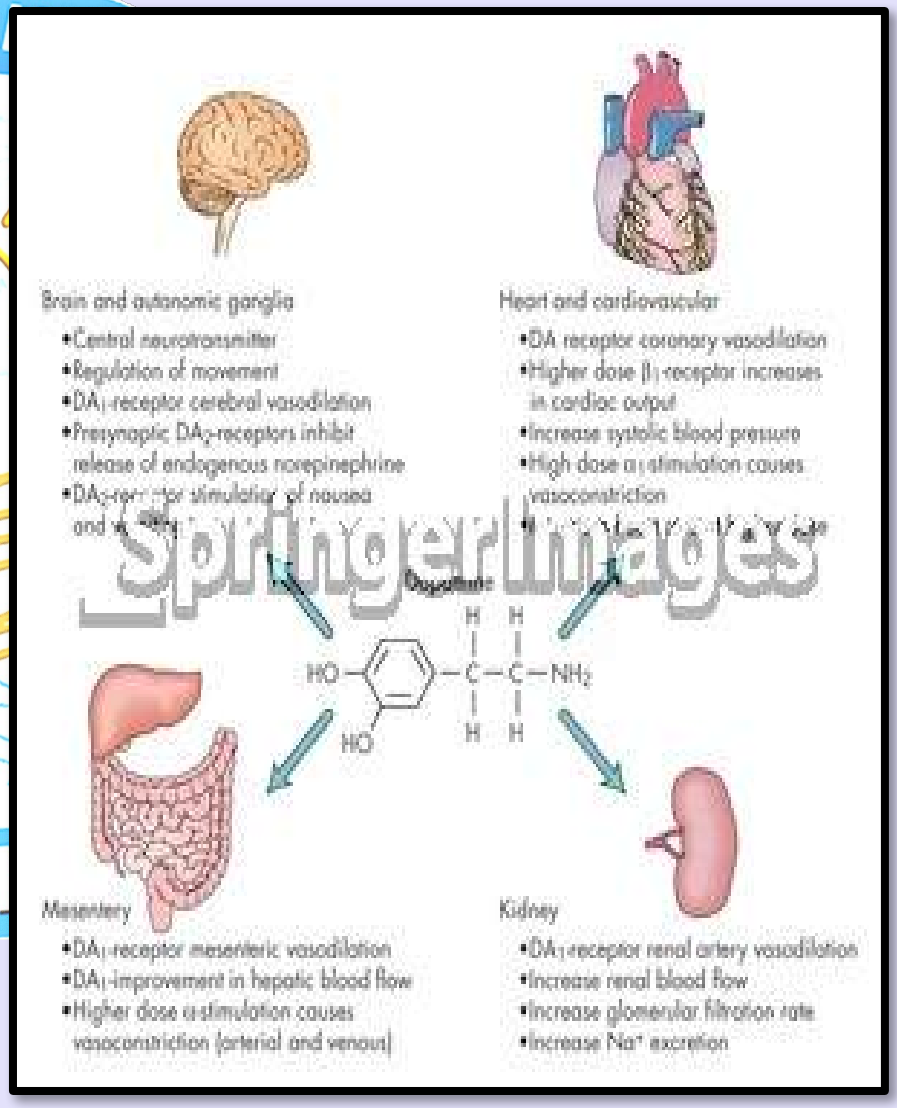
Norepinephrine

- Noradrenaline, also known as norepinephrine, comes from tyrosine,
- and is then broken down to form DOPA, which is broken down into dopamine.
- Approximately 15% of the noradrenaline produced is stored for later release.
- The remainder goes back to be broken down further into adrenaline.
- This will boost the supply of oxygen and glucose to the brain and muscles, and increase skeletal muscle readiness.
- This prepares your body to react to stressors.



DOPAMINE

- Dopamine, also known as DA, comes from noradrenaline and causes body to increase heart rate and blood pressure.
- It interacts with receptors on some nerve fibers and many neurons.
- Because it cannot go past the blood-brain barrier, given as a drug, it does not directly affect the central nervous system.
- Dopamine is not only a precursor of noradrenaline and adrenaline, but is also a neurotransmitter.
- It has many functions in the brain, including behavior and cognition, sexual gratification, mood, learning, sleep, memory, motivation, lactation, and involuntary movement.



Phaeochromocytoma

- Tumors secreting adrenaline/nor-adrenaline



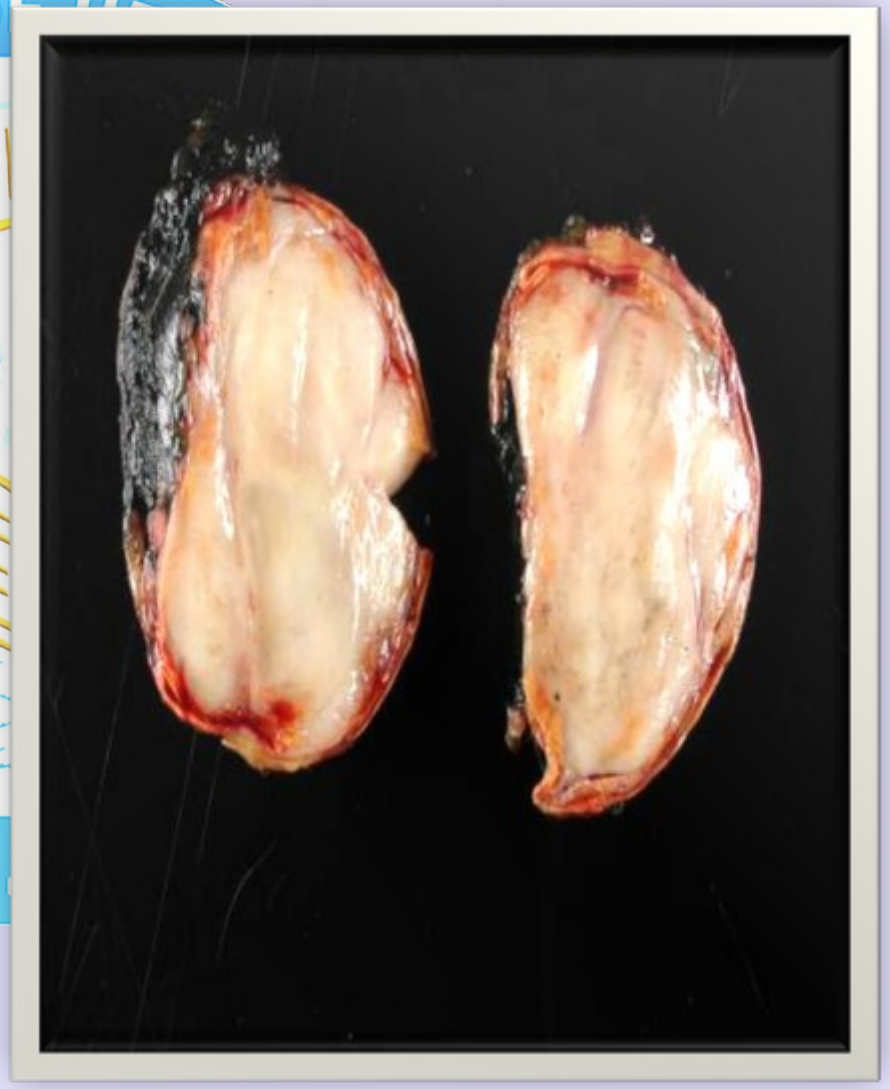
Neuroblastoma

- a neuroendocrine tumor of any neural crest tissue of the sympathetic nervous system



Ganglioneuroma

- a tumor in the nerve cells of the peripheral nervous system



THANKYOU